

Optimization

3.2 Derivative-free methods

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Why derivative-free methods

- **Black box:** f is the output of a simulation, of a test or of a code for which we only have the executable — there are values of f , there is no expression.
- **Noise:** if f comes from measurements or from simulations with tolerances, finite differences amplify the noise (they divide by h): the gradient approximation may be swamped by the noise and become unreliable.
- **Cost:** a gradient by central differences costs $2n$ evaluations; in many problems each evaluation takes minutes.
- **Nondifferentiability:** functions with $|\cdot|$, maxima, discrete decisions inside the model.

What derivative-free methods do

The methods we will see use **only evaluations of f** , building the search from the comparison of these values, without explicit gradients. They pay for it in speed: they must infer from the comparisons where the function decreases. **General rule:** numerical derivatives can amplify noise, and so these methods can be attractive; but noise also alters comparisons of values of f , so the methods presented are *not* automatically robust to noise.

The five methods of 3.2

Method	Idea	Section
Random search	evaluate f at random points; keep the best	3.2.1
Grid search	evaluate f at the nodes of a regular grid; keep the best (exploration, initialization)	3.2.2
Nelder–Mead simplex	$n + 1$ points; replace the worst by reflecting it through the others	3.2.3
Box's method	evaluate the 2^n vertices of a box around the point; move to the best, else shrink	3.2.4
Hooke–Jeeves	explore axis by axis; if it improved, jump in the direction of progress	3.2.5

For each one: what it uses, how it chooses where to evaluate, its cost in evaluations of f and its behaviour (for local methods, in the Rosenbrock valley). We start with the simplest (3.2.1–3.2.2) — random search is also the **baseline** against which the use of the same evaluation budget is compared.

Design of sandwich panels to reduce noise transmission (IDMEC)

Decision the design parameters of the panel.

Objective the vibroacoustic performance — computed by a *numerical model* of the panel.

Information on f values only: each evaluation is a simulation. In this problem, the model is used as a black box and does not provide gradients directly; obtaining them by finite differences would require $2n$ simulations per gradient, inheriting the numerical error of the model.

Method derivative-free direct search (DMS — the multiobjective version, 6.5, of the kind of methods in this section).

This is the typical case of a **real problem**: f is an expensive black box. The question “what information do we have about f ?” is answered before choosing the method — and here the answer makes the methods of 3.3, as presented, unattractive without an efficient way to obtain derivatives. (Textbook problem: Rosenbrock, where f , ∇f and $\nabla^2 f$ cost microseconds and the real cost is that of *understanding* the methods.) Next (3.2.1): the simplest — **random points**.

References Full list: *Course bibliography*, at the end of the course PDF.

- Conn, A. R., Scheinberg, K., Vicente, L. N., *Introduction to Derivative-Free Optimization*, SIAM (MPS-SIAM Series on Optimization 8), 2009 — Ch. 1.
- Kolda, T. G., Lewis, R. M., Torczon, V., *Optimization by direct search: new perspectives on some classical and modern methods*, SIAM Review 45(3):385–482, 2003.

Resources: DMS code in 6.5 (sandwich panel case).